Causal RL with Hidden Confounders and Partial Observability

A Literature Review

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Outline

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2. Key Concepts (the shared toolbox)
3. Map of the Literature
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Why this problem? Learning to act from logged data

The offline (batch) RL setting

In many high-stakes domains we **cannot experiment** — we only have a historical log of decisions and outcomes (e.g. ICU treatment records, recommendation logs). The goal: learn a **good policy** from this fixed dataset, without further interaction.

Two complications stacked together

- **Partial observability:** the true state is hidden; we see only noisy observations $\rightarrow$ a **POMDP**.
- **Hidden confounding:** the data-logging (behavior) policy acted on the **hidden state**, which also drives rewards and transitions.

Running example (sepsis treatment)

A clinician sees lab values but treats based on unrecorded judgment (severity, comorbidities). That hidden judgment shapes *both* the action *and* the outcome — the textbook recipe for a confounder.

The intuition: correlation that vanishes under intervention

A classic confounding fallacy

A town finds people who *carry lighters* get lung cancer more often.

$$\mathbb{P}(\text{cancer} \mid \text{lighter}) \text{ is high, but } \quad \mathbb{P}(\text{cancer} \mid \text{do}(\text{lighter})) = \mathbb{P}(\text{cancer} \mid \text{do}(\text{no lighter})).$$

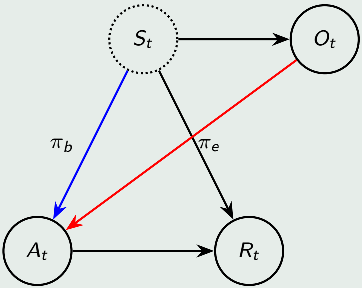
Smoking is the hidden confounder. Handing out lighters changes nothing.

Same trap in RL

A behavior policy that secretly conditions on the hidden state can make a **bad policy look good** (and vice-versa). Naively replaying the data mis-ranks policies. We need **Pearl's do($\cdot$)**, not raw correlations.

Why traditional RL / OPE fails here

The causal picture (one step)



Latent state S_t is a hidden common cause of R_t and A_t (via π_b), making it an unobserved confounder. The evaluation policy π_e operates only on the noisy proxy O_t and the observable history.

Consequences

- No Markov property on observations $\Rightarrow$ standard MDP off-policy evaluation (OPE) is **biased**.
- The bias does not shrink with more data — it is **structural**.
- Two open problems are coupled: *confounding bias* **and** *distribution shift* (the log under-covers good policies).

Problem statement

What all eight papers are really attacking

Given an **offline, confounded POMDP dataset** $\mathcal{D} = \{(O_t, A_t, R_t)\}$ generated by a behavior policy that depends on the *hidden* state, either

1. **Evaluate** a target policy's value V^π (Off-Policy Evaluation), or
2. **Optimize** — find a near-optimal $\hat{\pi}$ — *without* assuming full data coverage.

Our focus

The **core paper** (Hong et al., 2024) gives the first *model-based* answer with function approximation, and (Lu et al., 2024) gives the leading *model-free* rival. The other six papers form the surrounding toolbox that makes both legible.

Four ideas that recur across the literature

1. Proxy variables & proximal inference

We cannot see S_h, but **past and future observations** (O_{h-1}, O_h) are noisy "shadows" of it. Used as *negative controls*, they let us cancel the confounding bias without ever recovering S_h.

2. Bridge functions

Solutions to **conditional moment restrictions** that re-express an unidentifiable quantity using only observables, e.g.

$$\mathbb{E}\big[\mathrm{b}_R(O, A) \,\big|\, Z, A\big] = \mathbb{E}\big[R \,\big|\, Z, A\big].$$

3. Pessimism

Under partial coverage we act **cautiously**: maximize the *worst-case* plausible value. Guards against over-rating actions the data barely supports.

4. Minimax / spectral estimation

The moment restrictions become $\min - \max$ games (or moment-matrix algebra), enabling flexible function approximation in continuous spaces.

The landscape: eight papers, three axes

#	Paper	Goal	Deconfounding mechanism	Model-based
1	Hong et al. '24 **(core)**	Optimize	Reward/dynamic bridge functions	**Yes**
2	Lu et al. '24 **(P3O)**	Optimize	Bridge functions via minimax	No
3	Shi et al. '22	Evaluate	Minimax bridge (continuous)	No
4	Tennenholtz et al. '20	Evaluate	Proxies (first tabular result)	No
5	Bennett & Kallus '21	Evaluate	Proximal causal inference	No
6	Nair & Jiang '21	Evaluate	Spectral / moment matrices	No
7	Kallus & Zhou '20	Bounds	Sensitivity model $\Gamma \geq 1$	No
8	Lu et al. '18	Optimize	VAE latent model + do($\cdot$)	**Yes**

Most work is **evaluation**; only 1, 2, 8 reach **optimization**.
Only 1 & 8 are **model-based**. We anchor on the 1-vs-2 rivalry.

Key idea: don't estimate the value directly — learn the *models*

Extract the **reward model** and **transition model** from confounded data via two families of bridge functions solving conditional moment restrictions:

$$\mathbb{E}\big[b_R^{[t]}(A_t, O_t, r, o) \mid H_{t-1}, A_t, O_0\big] = p(r, o \mid H_{t-1}, A_t, O_0),$$

and an analogous *dynamic-emission* bridge $b_D^{[t]}$ for transitions.

Two-stage nonparametric estimation

Solve the moment restrictions in **RKHS** (kernel ridge), allowing general function approximation in continuous state/observation spaces.

Conservative (pessimistic) optimization

$$\hat{\pi} = \arg\max_{\pi \in \Pi} \min_{(b_R, b_D) \in \mathrm{conf}(\alpha, \beta)} V(\pi, b_R, b_D)$$

optimize the worst case over confidence regions of the bridge functions.

Paper 1 — what we get out of it

Guarantees

- Finite-sample suboptimality bound under **partial coverage** only (data must cover the optimal policy, not everything).
- Bound is **independent of the policy-class size** $|\Pi|$ — a key edge over model-free rivals.
- As a by-product it identifies the **full reward distribution** $p^{\pi}(r_t)$, not just its mean.

Caveats

- Relies on **completeness** + **existence of bridge functions** (untestable from data).
- Needs a baseline covariate O_0 (Assumption 3.1).
- **Purely theoretical** — no experiments at all.

"First model-based identification for confounded POMDPs with function approximation."

Paper 2 — P3O: a model-free rival

Proxy variable Pessimistic Policy Optimization

Skip modeling the environment. Use **proximal causal inference** to identify each policy's value through *value bridge functions* b^π obeying Bellman-like moment equations, estimated by **minimax**:

$$\hat{b}_h(b_{h+1}) = \arg\min_{b\in\mathcal{B}} \max_{g\in\mathcal{G}} \hat{\Phi}^{\lambda}_{\pi,h}(b, b_{h+1}; g).$$

Pessimism via coupled confidence regions

Build a sequence of confidence regions $\mathrm{CR}_\pi(\xi)$ around b^π, then

$$\hat{\pi} = \arg\max_{\pi} \min_{b\in\mathrm{CR}_\pi(\xi)} \hat{F}(b).$$

Guarantee

First provably efficient offline RL for confounded POMDPs:

$$\mathrm{SubOpt}(\hat{\pi}) = \widetilde{\mathcal{O}}\Big(H\sqrt{\tfrac{C^{\pi^\star}\log N_{\mathrm{fun}}}{n}}\Big) = \widetilde{\mathcal{O}}(n^{-1/2}).$$

Carries a $\sqrt{\log|\Pi|}$ factor.

Head-to-head: the central rivalry

	Paper 1 (model-based)	**Paper 2 / P3O (model-free)**
What is learned	reward & transition models	policy value directly
Estimator	two-stage RKHS / kernel ridge	min-max (minimax)
Deconfounding	emission bridge functions	value bridge functions
Pessimism	confidence regions on b_R, b_D	coupled regions $\mathrm{CR}_\pi(\xi)$
Rate	finite-sample, **free of $\lvert\Pi\rvert$**	$n^{-1/2}$ with $\sqrt{\log\lvert\Pi\rvert}$
Bonus	full reward *distribution*	weaker existence condition
Experiments	**none**	none

Same assumptions family, opposite philosophy — this is the comparison our project is built around.

Foundations & proxy-based OPE (Papers 4, 5, 3)

Paper 3 — Shi et al. '22

A **minimax learning** approach to confounded-POMDP OPE in continuous spaces — the function-approximation backbone Papers 1 & 2 build on.

Paper 4 — Tennenholtz et al. '20: the origin point

First **exact OPE** result for general POMDPs. Shows OPE in POMDPs is fundamentally different from MDPs, and that ignoring confounders rates *bad policies as good*. Uses past/future observations as proxies (tabular).

Paper 5 — Bennett & Kallus '21

Proximal RL: brings causal proxy variables into OPE, handling continuous spaces and bridging the hidden-confounder gap (sepsis-style tasks).

Alternative toolkits (Papers 6, 7, 8)

Paper 6 — Spectral (Nair & Jiang '21)

Identifies value with **moment matrices / spectral** algebra (HMM-style). Replaces the strict "$|\text{obs}| = |\text{state}|$" condition with a weaker *rank* condition. No model fitting, fewer causal assumptions.

Paper 8 — Deconfounding (Lu et al. '18)

The **historical on-ramp:** a VAE + LSTM reconstructs the hidden confounder, then Pearl's do($\cdot$) yields a deconfounded reward for Actor–Critic. Intuitive, but identifiability is not guaranteed.

Paper 7 — Sensitivity bounds (Kallus & Zhou '20)

When point identification is impossible, get a **guaranteed interval** instead of a number. A sensitivity knob $\Gamma \geq 1$ caps the confounding strength:

$$\Gamma \uparrow \Rightarrow \text{ wider, safer bounds.}$$

The "robustness mindset" — our richest source of fallback ideas.

Comparison along three dimensions

1. Evaluation vs. Optimization

Papers 3–7 stop at **OPE**; only Papers 1, 2, 8 close the loop to **policy optimization** — and 1, 2 do so via pessimism.

2. How is confounding removed?

- **Proxy / bridge functions** — Papers 1, 2, 3, 5
- **Explicit latent modeling** (VAE + do) — Paper 8
- **Spectral / moment matrices** — Paper 6
- **Sensitivity bounds** — Paper 7

3. Model-based vs. model-free

Stronger guarantees & interpretability (Paper 1) vs. directness & fewer moving parts (Paper 2). The theory predicts a **crossover** no one has tested.

The gap we can fill

Theory is rich; validation is missing

The core paper is **purely theoretical — no experiments** — and several identification assumptions are *untestable from data*. That is our mandate: turn the theory into a **working, empirically-tested system** and stress its assumptions.

Open questions

- **When does model-based beat model-free?** The $|\Pi|$-free bound (P1) vs. $\sqrt{\log|\Pi|}$ (P2) crossover is never tested.
- **What if identification silently fails?** No diagnostic, no graceful fallback.
- **The reward *distribution* is identified but unused** — a door to risk-sensitive / distributional RL.
- **Two notions of uncertainty** (finite-sample vs. confounding-strength Γ) are never unified.

Concrete implementation plan

Build

- Implement the core paper's **two-stage RKHS** bridge-function estimator + pessimistic max–min optimization (no public code exists).
- Reproduce on a small **synthetic confounded POMDP**.

Benchmark

A reusable **confounded-POMDP benchmark** (Pendulum / CartPole / sepsis-style) with controllable confounding strength and coverage.

Compare & probe

- **Head-to-head** Paper 1 vs. P3O on identical data; sweep $|\Pi|$, reward cardinality, coverage to locate the crossover.
- **Ablations** on O_0 quality and kernel/regularization choices.
- Test whether **spectral tools** (Paper 6) estimate the bridges more cheaply.

Goal: the first empirical validation of model-based causal RL for confounded POMDPs.

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Thank you

Questions & discussion